(c) Not equal to $m+n-1$
(d) Not equal to $m+n+1$

3. In a degenerate solution number of allocations is
(a) equal to $m+n-1$
(b) not equal to $m+n-1$
(c) less than $m+n-1$
(d) greather than $m+n-1$



4. The Penalty in VAM represents difference between the first _____
(a) Two largest costs
(b) Largest and Smallest costs
(c) Smallest two costs
(d) None of these

5. Number of basic allocation in any row or column in an assignment problem can be
(a) Exactly one (b) at least one
(c) at most one (d) none of these

6. North-West Corner refers to _____
(a) top left corner
(b) top right corner
(c) bottom right corner
(d) bottom left corner

7. Solution for transportation problem using _____method is nearer to an optimal solution.
(a) NWCM (b) LCM
(c) VAM (d) Row Minima

8. In an assignment problem the value of decision variable x_{ij} is _____.
(a) 1 (b) 0
(c) 1 or 0 (d) none of them

9. If number of sources is not equal to number of destinations, the assignment problem is called_____
(a) balanced (b) unsymmetric
(c) symmetric (d) unbalanced

10. The purpose of a dummy row or column in an assignment problem is to
(a) prevent a solution from becoming degenerate
(b) balance between total activities and total resources
(c) provide a means of representing a dummy problem
(d) none of the above

11. The solution for an assignment problem is optimal if
(a) each row and each column has no assignment
(b) each row and each column has atleast one assignment
(c) each row and each column has atmost one assignment
(d) each row and each column has exactly one assignment

12. In an assignment problem involving four workers and three jobs, total number of assignments possible are
(a) 4 (b) 3 (c) 7 (d) 12

13. Decision theory is concerned with
(a) analysis of information that is available
(b) decision making under certainty
(c) selecting optimal decisions in sequential problem
(d) All of the above

14. A type of decision –making environment is
(a) certainty (b) uncertainty
(c) risk (d) all of the above

Miscellaneous Problems

1. The following table summarizes the supply, demand and cost information for four factors S_1, S_2, S_3, S_4 shipping goods to three warehouses D_1, D_2, D_3

	D_1	D_2	D_3	Supply
S_1	2	7	14	5
S_2	3	3	1	8
S_3	5	4	7	7
S_4	1	6	2	14
Demand	7	9	18	

XII Std - Business Maths & Stat EM Chapter 10.indd 261 2/29/2020 11:45:22 AM